Observability Debt in Agent Evaluation: Toward a Telemetry Layer for AgenticOS

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Abstract

Public agent evaluation today is split across three disconnected infrastructures: leaderboards, trace corpora, and telemetry logs. None of the sources we examine records, in one place, the fields a resource-aware system layer would need to attribute cost: task, model, scaffold and policy configuration, outcome, token and cache usage, retries, latency, and billing. We study what this missing record costs, drawing on three public datasets. With the Holistic Agent Leaderboard (HAL), raw dollar-cost orderings do carry across some benchmark pairs (Spearman ρ=0.74–0.93) but are largely explained by a persistent displayed-label cost tendency (pooled R^2=82.0%); they weaken sharply once we adjust for success (mean ρ=0.59 against 0.84). With 977 task- and model-matched trajectory pairs from Open-SWE-Traces, scaffold choice shifts measurable execution burden at fixed task and model (OpenHands versus SWE-agent turn ratio 0.85, interval 0.82–0.87), and the common belief that failures are far more expensive is not borne out under exact task matching (our design has 80% power for premium fractions at or above 0.65; we observe 0.44–0.49). Across two independent telemetry corpora (8,058 and 1,017 sessions) we document the resource profile a scheduler would need but that no benchmark links to an outcome: a 93.7% mean cache-read share, tool-error cascades $4.29\times$ more likely after an error, and the top 1% of sessions holding 48.3% of input-token volume. The three studies complement rather than combine. Studies 1 and 2 show that raw-cost orderings and naive failure premiums mix up which workloads were attempted with how agents behaved, and Study 3 shows the execution variables hidden under those totals are themselves uneven, heavy-tailed, and state-dependent. From these observations we derive a joint telemetry schema spanning identity, execution, resources, control, and outcome, and argue it belongs at the operating-system layer, because the layer that allocates resources is the one that must observe what those allocations depend on.

1 Why This Belongs at the OS Layer

The agents we evaluate now keep state, call tools, retry, and carry context over long runs. Reasoning about their resource use, for scheduling, routing, budgeting, or oversight, needs the same unified record that an operating system keeps for ordinary programs: a schema that ties what ran, on what, under what policy, with what outcome, and at what resource cost. We call the growing inability of public evaluation infrastructure to keep that record across task, model, execution policy, outcome, and system events *observability debt*. In every public source we checked, this debt blocks any reliable attribution of observed resource differences to workload, model, scaffold, configuration, or behavior.

Why is this an operating-system concern rather than a bookkeeping fix for benchmarks? An agentic OS sits in a specific place in the stack. It sees *identity* (which task, model, scaffold, and policy are running), *execution* (which tools fire, what context grows), *resources* (tokens, cache, time, memory,

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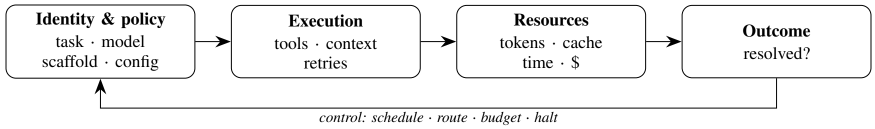

Figure 1: **The agentic OS control loop.** Each stage consumes exactly the schema fields derived in Section 6; the feedback arrow hosts scheduling, routing, budgeting, and halting decisions that today run on aggregate dollars because no examined public source closes this loop (Table 2).

money), *control* (retry, stop, route, budget), and *outcome* (did the task succeed). The same accounting logic that justifies OS-level resource management applies: the layer that runs and allocates must be able to see the telemetry those decisions require. Without per-round token detail, a memory manager cannot predict context growth. Without configuration identity, a router cannot learn *why* one task label is cheap. Without an error stream, a governor cannot stop a session that is spiraling. Today each of these choices is made indirectly from aggregate dollar totals (Figure 1).

We use three public datasets as stand-ins for what such a layer would unify: a cost-and-outcome leaderboard (HAL), a corpus of trajectories matched on task, model, and scaffold (Open-SWE-Traces), and raw session-level usage telemetry from two independent sources. Each shows only part of the picture. HAL links cost to outcome but hides the execution traces. Open-SWE-Traces gives exact task, model, and scaffold matching and a resolved or failed label, but no token or billing data. The telemetry corpora show token, cache, and tool dynamics, but no outcome or billing. **No public source we examined jointly exposes task, model, scaffold and policy configuration, outcome, token/cache usage, retries, latency, and billing**, the joint schema a resource-aware OS layer needs (Table 2). Putting the sources together does not close the gap: their differing execution units, identifiers, and release boundaries stop cost, outcome, and billing from being joined into one record per run.

Our three studies therefore complement rather than combine: each probes a different side of the gap, and none can be joined to another at the record level. (1) A portability study of cost on HAL shows that aggregate dollar cost mixes a displayed-label tendency with benchmark difficulty, and loses correlation once we adjust for success. (2) A matched-trajectory study shows that execution structure differs measurably across scaffolds, while a popular failure-cost rule in fact measures workload mix. (3) A cross-source telemetry audit shows that the execution variables hidden under aggregate cost are themselves uneven, heavy-tailed, and state-dependent. Together they point to one gap; no pair of them identifies a causal mechanism. This paper is an empirical requirements study for an AgenticOS telemetry layer, rather than an implementation of the layer itself.

2 Related Work

Kapoor et al. (2026) build HAL as cost-aware evaluation infrastructure covering coding, web, science, and customer-service tasks (21,730 rollouts, about $40,000 of evaluation spend). Kapoor et al. (2024) argue for reporting accuracy and cost together and for convex-hull frontier analysis, which we adopt as one of three frontier definitions. Ndzomga (2026) asks whether a subset of tasks preserves agent rankings within one benchmark; we ask instead whether a cost signal observed on one workload predicts cost on another. Benchmark^2 studies rank stability across static LLM benchmarks; our setting adds interactive scaffolds, dollar cost, success adjustment, and a held-out workload test. SWE-Effi reports that failed attempts use substantially more resources than successful ones; our matched design re-tests that claim at fixed task and finds no premium once task-difficulty confounding is removed (Section 4). Open-SWE-Traces provides the exact matching that makes this test possible, yet lacks token and billing fields, which motivates Section 6. Commercial tracers record per-deployment spans but are not built for cross-benchmark evaluation, and public releases strip execution detail, so the per-request visibility missing from our audited sources is exactly what we measure.

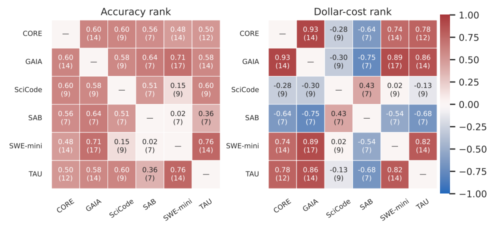

Figure 2: **Raw cost transfers; success-adjusted cost does not.** Pairwise Spearman ρ for benchmark pairs sharing ≥ 12 labels. Raw dollar cost is portable (mean 0.84); cost per expected success is not (mean 0.59), only three CPS pairs survive correction.

3 Study 1: Does a Cost Signal Travel Across Workloads?

3.1 Data and cohort

We use the public HAL CSV snapshot (Ndzomga, 2026): 242 displayed evaluations across nine benchmarks, 11 scaffolds, 29 labels, supplying accuracy and aggregate dollar cost but no token components, retries, or per-task trajectories—the gap this paper is about.

The only broadly repeated scaffold is the displayed HAL Generalist Agent; fixing it and matching exact label strings yields six adequately overlapping workloads (CORE-Bench Hard, GAIA, SciCode, ScienceAgentBench, SWE-bench Verified Mini, TAU-bench Airline), a cohort of 86 rows and 25 labels.

3.2 Raw cost transfers; the mechanism behind it does not

Across six pairs with ≥ 12 shared labels, raw dollar-cost Spearman ρ ranges 0.74–0.93 (mean 0.84), surviving Holm-corrected permutation testing (Figure 2). A pooled label-plus-benchmark log-cost fixed-effects model explains R^2=82.0% (adjusted 72.7%); once label and benchmark propensity are removed, **no high-overlap residual-cost association survives** a 20,000-draw permutation test.

Leave-one-benchmark-out (LOBO) prediction succeeds for CORE (ρ=0.78, N=14, p=0.0016), GAIA (0.87, N=16), SWE-mini (0.90, N=16), TAU (0.83, N=14). It fails for SciCode (unstable, $[-0.50, 0.14]$) and ScienceAgentBench (directionally negative, $[-0.77, -0.43]$).

Interpretation: the part of cost that does transfer behaves like a tendency tied to the displayed task label and persistent across workloads, plausibly set by provider pricing, reasoning-mode defaults, or routing policy, and in principle learnable by a router. Public data does not let us tell which, because the fields that would separate these explanations are not recorded.

3.3 Success-adjusted cost is a different, less portable ranking

Cost per expected success, $\mathrm{CPS}_{i,b}(\epsilon) = C_{i,b}/\max(A_{i,b}, \epsilon)$ with primary floor ϵ=0.01 (sensitivities at 0.001, 0.05, and zero-exclusion in the supplement), has mean ρ=0.59. Only three pairs survive correction: CORE–GAIA (0.87, $[0.63, 0.96]$), CORE–TAU (0.83, $[0.31, 1.00]$), GAIA–TAU (0.78, $[0.42, 0.94]$). The clearest raw-vs-CPS gap is GAIA–SWE-mini (0.60, $[0.14, 1.13]$): a label cheap in dollars is not reliably cheap per expected success elsewhere. No label tested on ≥ 5 workloads sits on every accuracy–cost frontier under discrete nondominance, convex hull, or 5% tolerance (o4-mini Low reaches $5/6$ under tolerance; no universal winner under any strict definition).

Implication: Routing on displayed dollar cost therefore optimizes a signal whose portability degrades substantially once success is taken into account.

4 Study 2: What Matched Trajectories Reveal That Aggregates Cannot

HAL's own trace archives are encrypted, so Study 1 cannot say *why* costs differ. We use an external unencrypted corpus—977 exact task–model pairs from Open-SWE-Traces for MiniMax-M2.5 under two scaffolds (OpenHands, SWE-agent), a reproducible but nonrandom four-shard boundary sample.

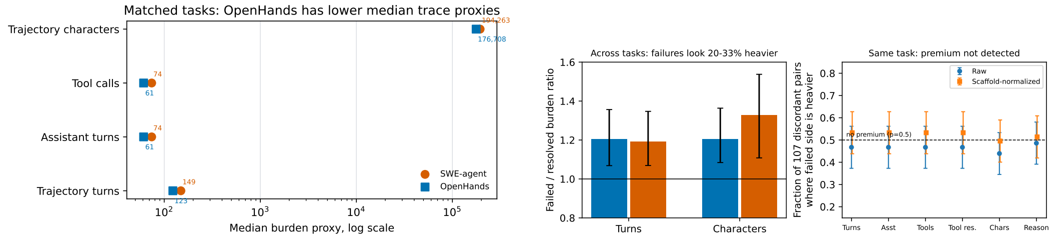

Figure 3: **(a)** Scaffold burden on 977 matched pairs: OpenHands uses $\sim$15% fewer median turns/tool calls than SWE-agent at fixed task and model. **(b)** The failure premium is not detected under exact task matching (80% power against $\geq$15pp). Left: naive cross-task ratios reproduce folklore (19–36% heavier failures). Right: among 107 discordant pairs, failed side heavier only 47–$52/107$.

Table 1: **Telemetry anatomy across two independent corpora.** Cache metrics are within-corpus; a share and a total-volume ratio are not interconvertible.

	TraceLab	MIMO Claude Code
Sessions / LLM rounds	8,058 / 665,453	1,017 / 4,690
Median input / output tokens per round	132,092 / 249	n/a (constant-tuple audit)
Cache-read vs. uncached input	93.7% mean share ($\approx$14.9$\times$)	753.0M / 46.3M tokens = 16.3$\times$
Median final-vs-initial context	3.6$\times$ (P90 9.9$\times$)	n/a
Top-1% / top-10% input-token share	48.3% / 88.2%	— / 37.9%
Error cascade P(err$\mid$err)/P(err$\mid$succ)	21.0/4.9% = 4.29$\times$	4.3% vs. 1.0% (4.33$\times$)
Errored calls retried same-tool / retry succeeds	88.3% / 77.9%	—
Consecutive-error run length (median/P90/max)	1 / 3 / 368	—
Tool-call latency, success vs. error (median)	140 ms / 313 ms	—
Tool-result size, success vs. error (median chars)	602 / 208	—

The schema exposes resolved status, roles, tool calls, and text but not billed tokens or dollars; we treat turns, tool calls, and characters as trace-burden proxies.

4.1 Scaffold choice changes observable burden at fixed task

OpenHands has lower matched medians: 123 vs. 149 trajectory turns (ratio 0.85, $[0.82, 0.87]$), 61 vs. 74 tool calls (0.85, $[0.82, 0.86]$), 176,708 vs. 194,263 characters (0.95, $[0.92, 0.98]$). Resolved status shows no detected difference (p=0.197): scaffolds differ in execution structure without a detected accuracy trade-off here.

4.2 The "expensive failure" premium is not detected under exact task matching

Comparing failed vs. resolved trajectories *across* tasks reproduces SWE-Effi's pattern (19–36% heavier failures). But among the **107 discordant pairs**—identical task and model, one scaffold resolved and the other failed—the failed side is heavier only 47–52 of 107 on every metric (binomial $p\geq 0.25$). A pre-specified power calculation gives 80% power against a premium fraction ≥ 0.65; the observed 0.44–0.49 does not resolve smaller effects ($<$ 15pp).

Interpretation: hard tasks make agents both spend more *and* fail more. A failure multiplier measured without conditioning on task therefore captures which tasks were attempted, not a property of failing runs themselves. Accounting that conditions on task needs exactly the joint record that no single public source provides.

5 Study 3: The Telemetry an OS Layer Would Need to See Directly

Studies 1–2 show aggregate cost hides *why* an agent is expensive and that execution structure varies across systems at fixed task and model. This section audits raw execution telemetry across two independent coding-agent corpora (Table 1). **TraceLab** Zhu et al. [2026]: $8{,}058$ sessions, $665{,}453$ rounds, 743,819 tool calls. **MIMO Claude Code** Zou [2026]: $1{,}017$ sessions, $4{,}690$ rounds. Neither links usage to outcome or billing.

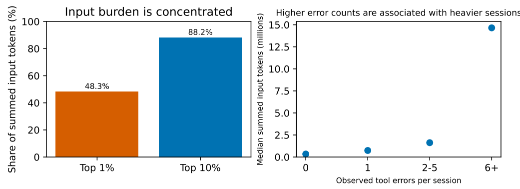

Figure 4: **Heavy-tailed burden.** Top-1%/top-10% summed-input concentration and median session input by observed-error bin (0.34M at zero to 14.65M at 6+).

Context is the dominant resource, not generation. The median TraceLab round carries 132,092 input tokens against 249 output tokens (93.7% mean cache-read share); median final-round context is 3.6× initial (P90 9.9×). A dollar total cannot separate price- from cache-policy- or accumulation-driven spend. Tool verbosity drives it: median next-input growth is 443 / 1,412 / 5,420 tokens below / within / above 10k tool-result characters.

Resource burden is heavy-tailed. The top 1% of sessions hold 48.3% of summed input volume; top 10% hold 88.2%. Median session input rises from 0.34M tokens at zero observed tool errors to 14.65M at six or more; mean-based budgeting would miss this tail-driven concentration (Figure 4).

Error cascades replicate across independent sources. After a successful call the next errs 4.9%, after an error 21.0% (4.29×); MIMO replicates at 4.33× (4.3% vs. 1.0%). Same-tool retry is common (88.3% get one, 77.9% succeed) but the tail is extreme (consecutive-error runs: median 1, P90 3, max 368). Failures are also slower yet less informative: 313 ms vs. 140 ms (P95 85.6 s vs. 15.8 s), returning 208 vs. 602 characters.

Data-quality boundary. MIMO repeats one constant per-round usage tuple, so its within-session token dynamics are not validated; we use it only for session counts, totals, concentration, and content-derived error sequences.

6 An Evidence-Derived Joint Telemetry Schema

Table 2 shows the coverage gap; the missing rows are exactly the ones Sections 3–5 show to matter. Each field below is tied either to an analysis above or to the OS control function it enables (the list is *evidence-derived*: it shows usefulness and necessity for those functions, not formal minimality). Grouped by function:

1. *(Attribution / exact matching)* **Task and model identity** (exact IDs, not display strings)—enables exact-matched designs; without it the Section 4 test is infeasible on public data.
2. *(Routing between scaffolds)* **Scaffold/harness identity and version**—attributes burden differences to scaffold design rather than model behavior; input to routing.
3. *(Configuration attribution)* **Configuration and policy identity** (reasoning mode, sampling params, budgets, routing/cache policy)—distinct from model identity; Study 1's unexplained propensity component is plausibly the shadow of these missing fields.
4. *(Success-conditioned accounting)* **Outcome/resolved status**—conditions any cost statistic on success, closing the CPS portability gap.
5. *(Memory / budget management)* **Token components per round (input, output, cache-read, cache-creation)**—separates price from accumulation and cache policy; lets a memory layer budget context growth.
6. *(Governance / halting)* **Tool-call counts, retries, error status per call**—detects and prices 4.29× cascade dynamics; a design motivation for governance policies that halt spiraling sessions—motivated here, not experimentally validated.
7. *(Real-time scheduling)* **Latency per call and per session**—present only as unlinked tool-call durations in one corpus; required for real-time scheduling.
8. *(Procurement / auditing)* **Billing, linked to all of the above**—the one field HAL exposes well, and exactly the field that is uninterpretable without the others.

Figure 1 places these fields at the stages of the control loop they close; Table 2 shows why, on public data alone, the loop does not close.

No public source we examined exposes this joint schema. We do not propose a storage format or API—a systems-design question outside our scope—but the measured consequences of its absence

Table 2: **The observability gap.** Coverage per public source examined: ■ exposed, ⊡ partial (display-label-only, degraded fidelity, or inferable), – absent. No source covers the joint schema; the union leaves latency unlinked and round-level token fidelity unverifiable outside one corpus.

Field	HAL	Open-SWE	TraceLab	MIMO
Task identity	⊡	■	–	–
Model identity	⊡	■	⊡	■
Scaffold identity/version	■	■	⊡	■
Configuration / policy identity	⊡	–	–	–
Outcome / resolved status	■	■	–	–
Tokens / cache components	–	–	■	⊡
Retries / error status	–	⊡	■	⊡
Latency	–	–	■	–
Billing	■	–	–	–

argue it belongs at a shared operating-system layer rather than being re-derived piecemeal inside each benchmark, where today it is not derived at all.

7 Limits

This work reuses existing data and has clear boundaries. Display labels do not fully identify prompts, tools, budgets, or harness versions, and the HAL cohort is small, uneven, and mostly single-run. Adjusting for label tendency is observational, not control of list price; CPS is an aggregate ratio, not the mean cost of successful runs. Study 2 uses one model on a non-random sample whose proxies are neither tokens nor dollars, and is powered only against large premiums ($\geq$ 15pp). Study 3 is observational and not linked to outcomes or billing; MIMO's per-round fields cannot be verified and are used only where supported, and its design implications are motivations, not tested policies. Each limitation is itself a case of the missing-field problem.

8 Conclusion

Across three studies, aggregate resource signals cannot be reliably attributed to workload, execution, or outcome without telemetry fields that the public sources we examined do not record together. Raw-cost orderings do carry across workloads, but they track a persistent displayed-label cost tendency more than they track inherent difficulty; success-adjusted cost reorders them (three of six pairs survive correction); and the assumed failure-cost premium does not appear under exact task matching. No source we examined links task, model, scaffold and configuration, outcome, token/cache usage, retries, latency, and billing in one place. The layer that runs execution and allocates resources is the natural owner of this telemetry. We therefore argue the schema belongs at a shared operating-system layer for agentic AI, rather than being rebuilt separately inside each leaderboard or trace release.

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